elemental copper is more stable than $Cu^{2+}$. There are two deviations., In the general trend, Fig shows that $\left(E^{0}_{M^{2+}/M}\right)$ value for manganese and zinc are more negative than the regular trend. It is due to extra stability which arises due to the half filled $d^5$ configuration in $Mn^{2+}$ and completely filled $d^{10}$ configuration in $Zn^{2+}$.

Transition metals in their high oxidation states tend to be oxidizing . For example, $Fe^{3+}$ is moderately a strong oxidant, and it oxidises copper to $Cu^{2+}$ ions. The feasibility of the reaction is predicted from the following standard electrode potential values.

$Fe^{3+}(aq) + e^{-} \rightleftharpoons Fe^{2+} \quad E^{0} = 0.77V$

$Cu^{2+}(aq) + 2e^{-} \rightleftharpoons Cu(s) \quad E^{0} = +0.34\ V$

The standard electrode potential for the $M^{3+}/M^{2+}$ half-cell gives the relative stability between $M^{3+}$ and $M^{2+}$. The reduction potential values are tabulated as below.

| Reaction | Standard reduction potential ( V ) |
|---|---|
| $Ti^{3+} + e^{-} \longrightarrow Ti^{2+}$ | –0.37 |
| $V^{3+} + e^{-} \longrightarrow V^{2+}$ | –0.26 |
| $Cr^{3+} + e^{-} \longrightarrow Cr^{2+}$ | –0.41 |
| $Mn^{3+} + e^{-} \longrightarrow Mn^{2+}$ | +1.51 |
| $Fe^{3+} + e^{-} \longrightarrow Fe^{2+}$ | +0.77 |
| $Co^{3+} + e^{-} \longrightarrow Co^{2+}$ | +1.81 |

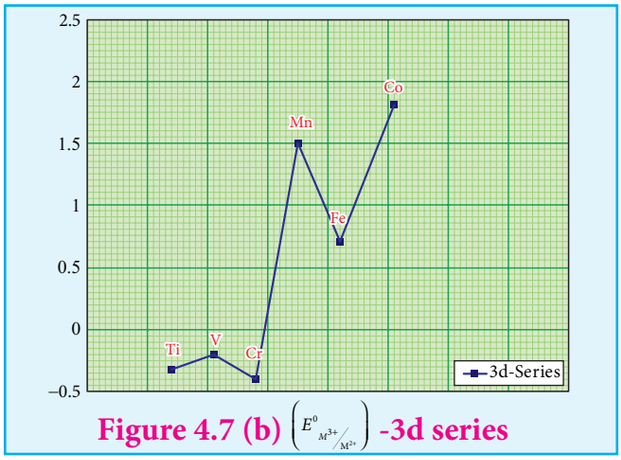


Figure 4.7 (b) $\left(E^{0}_{M^{3+}/M^{2+}}\right)$ -3d series

The negative values for titanium, vanadium and chromium indicate that the higher oxidation state is preferred. If we want to reduce such a stable $Cr^{3+}$ ion, strong reducing agent which has high negative value for reduction potential like metallic zinc $\left(E^{0} = -0.76\ V\right)$ is required.

The high reduction potential of $Mn^{3+}/Mn^{2+}$ indicates $Mn^{2+}$ is more stable than $Mn^{3+}$. For $Fe^{3+}/Fe^{2+}$ the reduction potential is 0.77V, and this low value indicates that both $Fe^{3+}$ and $Fe^{2+}$ can exist under normal conditions. The drop from Mn to Fe is due to the electronic structure of the ions concerned.$Mn^{3+}$ has a $3d^4$ configuration while that of $Mn^{2+}$ is $3d^5$. The extra stability associated with a half filled d sub shell makes the reduction of $Mn^{3+}$ very feasible ($E^{0} = +1.51V$).

### 4.3.6 Magnetic properties

Most of the compounds of transition elements are paramagnetic. Magnetic properties are related to the electronic configuration of atoms. We have already learnt in XI STD that



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